

Forecasting Climate Change Impacts on the Habitat of the American Pika

Introduction

Climate change is one of the greatest threats to global biodiversity, altering species distributions, ecosystem functions, and the availability of suitable habitats worldwide. Rising temperatures, changes in precipitation patterns, and increasing frequency of extreme weather events are forcing many species to shift their geographic ranges or adapt to rapidly changing environmental conditions (IPCC, 2023). **Predicting how species will respond to these changes has therefore become a major focus of conservation biology and ecological research. Species Distribution Models (SDMs), which combine species occurrence data with environmental variables, provide a powerful data-driven approach for identifying current habitat suitability and forecasting future changes under different climate scenarios** (Elith & Leathwick, 2009).

This project applies data science techniques to model the current and future habitat suitability of the American pika (*Ochotona princeps*). Using presence and absence records of the species together with a suite of climatic and environmental predictors—including temperature, precipitation, elevation, slope, and vegetation-related variables, like severe drought indexes—we develop predictive habitat suitability models. Future climate projections are then incorporated to estimate how suitable habitat may change under projected climate change scenarios, providing insights into potential range shifts, habitat fragmentation, and potential population declines.

The American pika (*Ochotona princeps*) is a small herbivorous mammal belonging to the family Ochotonidae, which is closely related to rabbits and hares. It inhabits alpine and subalpine mountain ecosystems throughout western North America, typically at elevations above 2,000 m, although some populations occur at lower elevations where local climatic conditions remain sufficiently cool (Smith & Weston, 1990). Pikas are strongly associated with talus slopes—accumulations of broken rock formed at the base of cliffs and mountain ridges—which provide shelter from predators, nesting sites, and access to cool underground microclimates. These rocky habitats contain numerous crevices that allow pikas to retreat below the surface during periods of extreme temperatures, creating a relatively stable thermal environment compared with the surrounding air.

Unlike many alpine mammals that hibernate during winter, pikas remain active throughout the year. Their survival depends heavily on a remarkable behavioural adaptation known as **haying**. During the short alpine summer, pikas spend much of their time collecting grasses, wildflowers, sedges, shrubs, and other vegetation, which they carefully dry in the sun before storing in large "haypiles" beneath rocks.

These food caches provide the resources needed to survive the long winter months when vegetation is unavailable (Smith & Weston, 1990). Because they rely on this brief growing season to accumulate sufficient food reserves, any climate-driven changes that reduce plant productivity or alter the timing of snowmelt and vegetation growth may significantly affect their survival and reproductive success.

American pikas also exhibit strong territorial behaviour. Individuals typically defend relatively small home ranges centred around their talus habitat and haypiles, using distinctive vocal calls to warn intruders and communicate with neighbouring individuals. Their limited dispersal ability means that movement between suitable habitat patches is often difficult, particularly when mountain habitats become increasingly fragmented. As climate change forces suitable environmental conditions to shift toward higher elevations, many pika populations may become isolated on mountain summits, where opportunities for further range expansion no longer exist—a phenomenon often described as the "escalator to extinction" (Beever et al., 2011).

The American pika has become an important indicator species for studying the ecological impacts of climate change because of its exceptional sensitivity to temperature. Pikas possess a high metabolic rate and dense insulating fur that enables survival during harsh winters but makes them vulnerable to overheating during warm weather. Studies have shown that prolonged exposure to temperatures above approximately 25–28°C can cause severe physiological stress and may even be fatal if individuals cannot access cooler refuges (Beever et al., 2011). Consequently, increasing temperatures, reduced snowpack, altered precipitation patterns, and more frequent heat waves threaten both the availability and quality of suitable habitat across their range.

An important aspect of pika ecology, however, is that talus habitats can partially buffer the effects of climate change. Air circulates through the network of spaces between rocks, producing cooler and more stable subsurface temperatures than those measured above ground. These cool microclimates allow pikas to remain active beneath the rocks during periods of high surface temperatures and may provide temporary refuges from regional warming (Millar & Westfall, 2010). In addition, winter snowpack acts as an insulating layer over the talus, protecting pikas from extreme cold temperatures. **Nevertheless, while these microrefugia may delay the impacts of climate change, they cannot fully compensate for long-term increases in regional temperatures or continued habitat loss. As warming progresses, the protective capacity of talus habitats is expected to diminish, making accurate predictions of future habitat suitability increasingly important.**

These ecological characteristics make the American pika an excellent model species for habitat suitability modelling. Its close association with specific environmental conditions, well-documented climatic sensitivity, relatively restricted dispersal, and

extensive occurrence records make it particularly suitable for evaluating how changing climatic conditions may alter species distributions. Furthermore, because pikas occupy alpine ecosystems that are themselves highly vulnerable to climate change, **understanding their future distribution provides valuable insights into the broader impacts of warming on mountain biodiversity.**

Beyond its ecological significance, this project contributes directly to several United Nations Sustainable Development Goals (SDGs). Most notably, it supports **SDG 13: Climate Action** by improving understanding of the ecological consequences of climate change and providing scientific evidence that can inform adaptation and conservation strategies. The project also contributes to **SDG 15: Life on Land**, which seeks to protect terrestrial ecosystems, halt biodiversity loss, and conserve mountain ecosystems. By identifying areas likely to remain suitable for pikas under future climate conditions, this study can help prioritize conservation actions, guide habitat management, and support biodiversity monitoring in vulnerable alpine landscapes.

From a data science perspective, this project demonstrates the application of machine learning and predictive modelling to real-world environmental challenges. Integrating species occurrence data with multiple climatic and environmental predictors illustrates how spatial datasets can be combined to generate evidence-based ecological forecasts. Such approaches are increasingly important for supporting conservation planning, informing policy decisions, and improving our understanding of how biodiversity may respond to future climate change.

Value proposition

The proposed data product is an interactive web-based platform that transforms complex species distribution modelling results into an accessible tool for education, exploration, and environmental awareness. Users can learn about the ecology of the American pika, understand how climate change affects both the species and alpine ecosystems, and interactively explore maps showing current habitat suitability alongside projected distributions under multiple future climate change scenarios. By presenting scientific predictions through intuitive visualizations, the platform helps users better understand the relationship between climate change and biodiversity, making ecological research more accessible to students, educators, researchers, conservation practitioners, policymakers, and the general public.

The platform addresses the growing need for accessible climate change communication by bridging the gap between advanced ecological modelling and public understanding. While scientific studies often present results in technical publications that are difficult for non-specialists to interpret, this web interface enables users to explore the data themselves, compare different climate scenarios,

and visualize how habitat suitability may change over time. This promotes greater awareness of the consequences of climate change, highlights the vulnerability of mountain ecosystems, and encourages informed discussions about conservation and climate action.

A key differentiator of the product is its integration of machine learning–based species distribution models with multiple environmental datasets—including climate variables, topographic characteristics, vegetation indices, and drought indicators—within an interactive geographic information system. Rather than providing static maps or descriptive summaries, the platform allows users to dynamically compare current and future habitat suitability under different climate projections, offering a more engaging and informative experience. In addition, the combination of ecological background information with predictive spatial analysis demonstrates the practical application of data science to real-world environmental challenges.

The value of the platform is supported by a strong scientific foundation. Habitat suitability predictions are generated using documented American pika occurrence records and environmental variables, following widely accepted ecological modelling approaches (Elith & Leathwick, 2009). The future projections are based on climate scenario data consistent with assessments by the IPCC (2023), while the ecological interpretation is supported by extensive research on the species' climate sensitivity (Beever et al., 2011; Millar & Westfall, 2010). By combining validated scientific data, predictive modelling techniques, and interactive visualization, the platform provides a credible and engaging resource that demonstrates the importance of data-driven conservation planning while increasing public understanding of the ecological impacts of climate change.

References

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